PUSHKAR VERMA

AI Engineer — Agent Systems & Production LLM Infrastructure

Mohali, India · Seeking relocation to the Netherlands · Eligible for the IND highly skilled migrant permit under the under-30 salary criterion (€4,357/month gross, excl. holiday allowance)

pushkar3698@gmail.com · +91 97792 60517 · github.com/pushkarverma3698 · linkedin.com/in/pushkar-verma-809155234

> SOURCE OF TRUTH. Mirrored from the Google Drive master
> ("Pushkar Verma — AI Engineer CV (MASTER)", file id
> 1-X6kH15txvru1drxcqWGc9-Wnn4TLViV7pLAJp2SlA8), last revised 29 July 2026.
> This file — not the synthesized wiki — is what `cv_gaps` compares the market
> against. The wiki is generated from chat transcripts and had garbled the
> Contact_ME dates and job title; a gap report is only as honest as the CV under it.

SUMMARY

I build agent systems that are safe to put in front of real users, and I can show the receipts. I designed, deployed and operate **FounderOS**, a LangGraph multi-agent kernel that takes real business actions — email, LinkedIn, GitHub, sandboxed shell jobs — behind human approval, running in production on its own VPS. The part that matters is not that it works: it is **structurally unable to claim an action it did not take**, its plans are reproducible at temperature 0, and 1,800+ tests run the full graph on every commit at $0. Three-plus years of production full-stack engineering underneath — React and Next.js interfaces, Node APIs, PostgreSQL, Linux.

SELECTED WORK

FounderOS — Production Multi-Agent Kernel

LangGraph JS (StateGraph) · TypeScript strict · Gemini Flash · PostgreSQL + Drizzle · Docker · systemd · GitHub Actions — github.com/pushkarverma3698/FounderOS

- **Zero-hallucination action layer.** Every tool call emits a code-written ToolReceipt; the response synthesizer only ever sees validated results. The agent cannot report an action that did not happen — enforced by the type system and the graph, not by prompt instructions.
- **Deterministic orchestration.** The supervisor is pure code, not an LLM guessing at routing: typed plan → dispatch → isolated worker → Zod-validated StepResult. CI runs the golden set twice and fails if the two plans differ.
- **Crash-safe human-in-the-loop.** The approval row is written to Postgres *before* interrupt() fires, and an idempotency check runs before every external send — so a crash mid-approval or a retried message can neither lose an approval nor double-send an action.
- **Three-tier evaluation.** 1,800+ deterministic kernel tests at $0 per run on every commit; a 29-task golden eval scoring ~90% routing accuracy, ~96% tool selection, ~89% HITL coverage; and a 22-task live acceptance run before deploy.
- **Architecture discipline enforced in CI.** Five automated fitness functions — tombstones on deleted modules, an architecture-debt ratchet that may only shrink, import-direction checks, a 400-line file budget, and mandatory tagging on fail-open code — to stop AI-assisted code quietly bloating into something unmaintainable.
- **Genuinely in production.** systemd service with crash-loop breaker, Postgres/pgvector and a local LLM under Docker Compose, a watchdog health-checking every 2 minutes, and a CI/CD pipeline that only restarts the live service if every step passes — a failed deploy leaves the previous version running.
- **Sandboxed execution.** vps_run executes a single shell command inside an ephemeral, network-isolated Docker sandbox: pinned image allowlist, no network by default, CPU/memory/process caps, hard timeout,

and an explicit guard refusing to mount the live app's own source. Idempotent under checkpoint replay; artifacts harvested to S3 regardless of exit code.

Agent Reliability Benchmark — open harness *(in progress)*

Measuring what agent portfolios assert: fabricated-action rate, plan determinism, cost and failure-recovery behaviour across three orchestration strategies — a contract-first kernel, a naive ReAct loop, and raw tool-calling — on the same model, temperature, tools and task set. Ground truth comes from code-recorded receipts rather than an LLM judge, so the grader itself cannot hallucinate. Harness, raw transcripts and a threats-to-validity section published for dispute.

Turicks 3D — Cinematic Interactive Web Experience

Next.js 16 · React 19 · Motion 12 · custom Canvas engine · TypeScript — live: main.dgr0xigg9rbz4.amplifyapp.com

- Profiled before optimising: the stutter on mid-range Android was WebGPU/React-Three-Fiber shader rendering, not asset size. Built a custom ImageSequenceEngine pre-decoding 1440p frames off the main thread onto an OffscreenCanvas — locked 60fps on both iOS and Android.
- Deliberately skipped the WebGPU/3D-geometry stack most 2026 portfolio sites default to: the goal was cinematic scroll storytelling over photography, which does not need real 3D geometry. Canvas plus decoded frames was the simpler, more reliable bet.
- Adaptive resolution tiering selected from device memory and bandwidth, plus an accessible flat fallback for reduced-motion users — built in from day one, not retrofitted.

EXPERIENCE

Turicks — Software Engineer (self-employed) | Mar 2026 – Present | Remote

- Designed, deployed and operate FounderOS in production (above), handling day-to-day operations behind a Telegram approval gate.
- Built turicks.com end to end on Next.js App Router, TypeScript and Tailwind CSS — marketing, blog and case studies, inside a 3–5 day build cycle.
- Built real-time operator dashboards (Next.js, LangGraph JS, PostgreSQL) showing live agent run timelines and streaming output, used to monitor and debug FounderOS in production.
- Delivered full-stack builds for EdTech, healthcare and SaaS clients — Figma through React frontends and Node API routes, deployed on Vercel and AWS Amplify.

Contact_ME — Full Stack Engineer | Nov 2025 – Mar 2026 | Gurugram, India (Hybrid)

- Sole frontend owner: built a reusable React component library from the Figma design system, cutting time-to-build a new screen from roughly two days to under one.
- Cut initial page load ~35% (Lighthouse low-60s → high-80s) via route-splitting and deferring non-critical assets.
- Integrated and debugged REST APIs, owned auth and session state, and fixed backend issues directly — turning 2–3 day backend-queue bugs into same-day fixes.
- Deployed and monitored on Microsoft Azure (App Service, DevOps pipelines) alongside Vercel and AWS Amplify, managing environment config and rollbacks.

HFN — Associate Software Developer | Jul 2023 – Jun 2025 | Mohali, India (Hybrid) *Joined Apr 2023 as Software Engineer Intern.*

- Built and maintained React.js + Material-UI admin dashboards — leaderboard, task progress, calling queue, summary screens — used daily by a 50-person operations team.
- Wrote and maintained 25+ Node.js/Express REST endpoints powering those dashboards, designing interface and data model together.
- Replaced a manual weekly reporting process with one-click PDF generation built into the dashboard — a ~3-hour/week task reduced to seconds, roughly 150+ hours a year returned to the ops team.

- Contributed frontend to HFN Saarthi, an Android app for farmers, published on the Google Play Store.

AlmaBetter — Full-Stack Developer Apprenticeship | Jan 2022 – Jan 2023 | Remote

- Year-long apprenticeship shipping React, Node.js and MongoDB projects under mentorship and structured code review — the transition from a humanities degree into software engineering.

SKILLS

AI / Agent Systems — LangGraph JS, multi-agent orchestration, LLM tool-calling, human-in-the-loop design, Zod contract validation, eval harness design, LangSmith tracing, RAG and hybrid retrieval, MCP (Model Context Protocol), Gemini API, OpenRouter, prompt engineering, token/cost budgeting

Backend & Data — Node.js, TypeScript (strict), Express, REST API design, PostgreSQL, pgvector, Drizzle ORM, MongoDB, Redis, authentication and sessions

Infrastructure & Ops — Linux VPS administration, Docker & Docker Compose, systemd, GitHub Actions CI/CD, sandboxed job execution, health-check and watchdog design, observability, Postgres backup/restore

Frontend — React, Next.js (App Router), TypeScript, Tailwind CSS, Material-UI, Canvas/WebGL engines, React Three Fiber, GSAP/Motion, performance optimisation (CLS, 60fps, Lighthouse), responsive and accessible layouts

Cloud — AWS (EC2, S3, CloudFront, CloudWatch, Amplify), Azure (App Service, DevOps Pipelines), Cloudflare (Pages, Workers), Vercel, Hetzner

EDUCATION

Bachelor of Arts — Panjab University, Chandigarh | Dec 2018 – May 2022